Diffusion of the Philips Norelco i9000 Prestige Ultra

DS 223 Marketing Analytics - Homework 1: Bass Model

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1 Summary

This report predicts how the **Philips Norelco i9000 Prestige Ultra**, a premium smart electric shaver, could spread in **France**. It uses the Bass model. The model learns how fast people adopt (p and q) from the past sales of men's razors and trimmers in France. The size of the market (M) for the i9000 is estimated with simple Fermi logic.

2 Step 1: The Innovation

I chose the **Philips Norelco i9000 Prestige Ultra** from TIME's The Best Inventions of 2025 list.

"This electric shaver introduces SenseIQ Pro, a new version of Philips Norelco's AI-backed technology that helps users find the right shaving pressure via LED guidance lights (orange means too hard, green is just right). Also new is the i9000's Triple Action Lift & Cut system, which lifts hair from its root to cut it just below the skin's surface, up to -.08mm. Flexible heads adapt to facial curves to shave hard-to-reach areas, different modes handle everything from thick beards to foam, and, in another first, the charging station's hood sanitizes the shaver with UV light while the battery refills." (Retrieved from here)

3 Step 2: A Similar Innovation from the Past

- **Function:** both remove facial hair for the same buyers.
- **Technology:** electric razors and trimmers use a motor and blades. The i9000 adds sensors and AI to adjust the shave.
- **Market impact:** the i9000 is a premium upgrade of a product people already know, so its spread should look like the spread of electric grooming devices.
- **Look-Alike:** The look-alike is the earlier generation of men's electric grooming devices.

4 Step 3: Historical Data

Source: Statista (2024). Number of men's grooming products sold annually in France from 2010 to 2023 (in 1,000 units). Data from GIFAM, published September 2024. Statista ID 747972. Check the source here

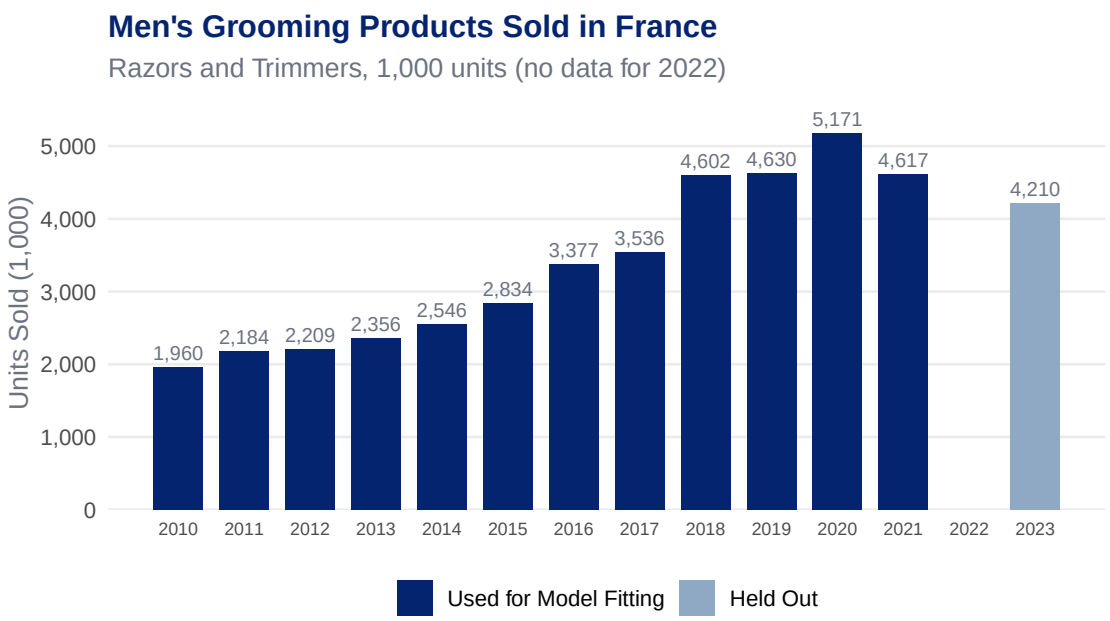
The data covers razors and beard/body trimmers. The source has no data for 2022. The 2023 value is not used to fit the model because it is an estimate. I keep it as a check.

```
grooming <- read.csv("data/men_grooming_france.csv")
grooming$t <- grooming$year - 2009 # t = 1 is 2010
grooming$used <- ifelse(grooming$year <= 2021, "Used for Model Fitting", "Held Out")
grooming$used <- factor(grooming$used, levels = c("Used for Model Fitting", "Held Out"))
kable(head(grooming, 3))
```

year	sales	t	used
2010	1960	1	Used for Model Fitting
2011	2184	2	Used for Model Fitting
2012	2209	3	Used for Model Fitting

```
fig_hist <- ggplot(grooming, aes(x = year, y = sales, fill = used)) +
  geom_col(width = 0.75) +
  geom_text(aes(label = comma(sales)), vjust = -0.5, size = 3,
            colour = col_grey) +
  scale_fill_manual(values = c("Used for Model Fitting" = col_main,
                               "Held Out" = col_light)) +
  scale_x_continuous(breaks = 2010:2023) +
  scale_y_continuous(labels = comma, expand = expansion(mult = c(0, 0.1))) +
  labs(title = "Men's Grooming Products Sold in France",
       subtitle = "Razors and Trimmers, 1,000 units (no data for 2022)",
```

```
       x = NULL, y = "Units Sold (1,000)") +
  theme_bass() +
  theme(axis.text.x = element_text(size = 8))
fig_hist
```




```
ggsave("img/fig1_historical_sales.png", fig_hist,
       width = 8, height = 4.5, dpi = 300)
```

Sales grew every year from 2010 to 2020 and then fell. This is the shape the Bass model needs. It grows, reaches a peak, and starts to decline.

5 Step 4: Estimating p, q, and M

I estimated the parameters in two ways: Non-Linear Least Squares (`nls()`) and the `diffusion` package.

```
# General Formula
bass_formula <- sales ~ m*(((p+q)^2/p)*exp(-(p+q)*t))/(1+(q/p)*exp(-(p+q)*t))^2

fit_data <- grooming[grooming$year <= 2021, ]
sales <- fit_data$sales
t <- fit_data$t

bass_model <- nls(
  bass_formula, data = fit_data,
  start = c(list(m =sum(sales)),p = 0.02, q = 0.4))
kable(round(summary(bass_model)$coefficients, 4))
```

	Estimate	Std. Error	t value	Pr(>|t|)
m	132297.6527	64934.9540	2.0374	0.0721
p	0.0113	0.0044	2.5442	0.0315
q	0.1471	0.0358	4.1061	0.0027

```
diff_m <- diffusion(sales)
diff_m
```

```
## bass model
##
## Parameters:
##      Estimate p-value
## m 146335.0282      NA
## p      0.0113      NA
## q      0.1372      NA
##
## sigma: 281.7347
```

```
m <- coef(bass_model)[["m"]]
p <- coef(bass_model)[["p"]]
q <- coef(bass_model)[["q"]]

t_star <- log(q / p) / (p + q)
data.frame(
  Peak_t = round(t_star, 1),
  Peak_year = round(2009 +t_star),
  Actual_peak_year = grooming$year[which.max(grooming$sales)])
```

```
##   Peak_t Peak_year Actual_peak_year
## 1   16.2      2025             2020
```

5.1 Results

- **p = 0.0113** (innovation): a very small share of people buy on their own, without being influenced by others.
- **q = 0.1471** (imitation): about 13 times larger than p. Most growth comes from people copying other buyers.
- **M = 132,298** thousand units (about 132 million units): the total number of units the model expects in this market over its whole life. This value is very uncertain (its standard error is about half of its size).
- The `diffusion` package gives similar numbers: p = 0.0113, q = 0.1372, M = 146,335. The small differences come from the different optimization methods.

5.2 Checking the Model

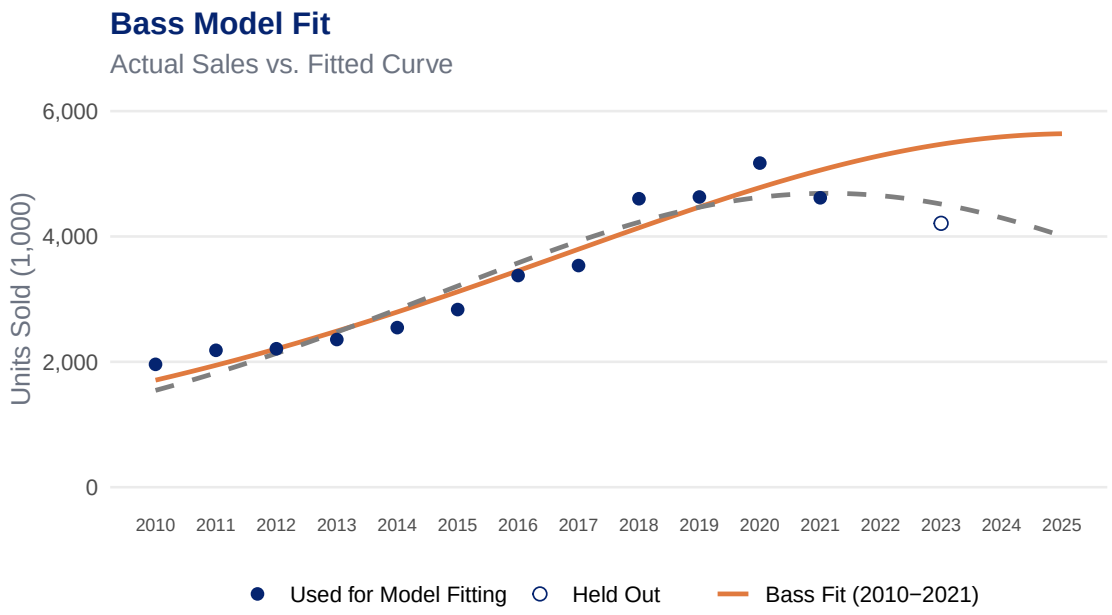
```
bass_all <- nls(
  bass_formula, data = grooming,
  start = c(list(m = sum(grooming$sales), p = 0.02, q = 0.4)))
```

```r
res <- data.frame(
  Model = c("2010-2021 (main)", "2010-2021 and 2023"),
  M = c(coef(bass_model)[["m"]], coef(bass_all)[["m"]]),
  p = c(coef(bass_model)[["p"]], coef(bass_all)[["p"]]),
  q = c(coef(bass_model)[["q"]], coef(bass_all)[["q"]]))
kable(res, digits = c(0, 0, 4, 4), row.names = FALSE, format.args = list(big.mark = ","))
```

Model	M	p	q
2010-2021 (main)	132,298	0.0113	0.1471
2010-2021 and 2023	83,727	0.0155	0.1917

```r
curve_df <- data.frame(t = seq(1, 16, by = 0.1))
curve_df$year <- curve_df$t + 2009
curve_df$main <- bass.f(curve_df$t, p, q) * m
curve_df$all  <- bass.f(curve_df$t, coef(bass_all)[["p"]],
                        coef(bass_all)[["q"]]) * coef(bass_all)[["m"]]

fig_fit <- ggplot() +
  geom_line(data = curve_df, aes(x = year, y = main,
            colour = "Bass Fit (2010-2021)"), linewidth = 1) +
  geom_line(data = curve_df, aes(x = year, y = all,
            colour = "Bass Fit (2023 included)"), linewidth = 1,
            linetype = "dashed") +
  geom_point(data = grooming, aes(x = year, y = sales, shape = used),
             size = 2.5, colour = col_main) +
  scale_colour_manual(values = c("Bass Fit (2010-2021)" = col_accent,
                                 "Bass Fit (with 2023)" = col_grey)) +
  scale_shape_manual(values = c("Used for Model Fitting" = 16,
                                "Held Out" = 1)) +
  scale_x_continuous(breaks = 2010:2025) +
  scale_y_continuous(labels = comma, limits = c(0, 6000)) +
  labs(title = "Bass Model Fit", subtitle = "Actual Sales vs. Fitted Curve",
       x = NULL, y = "Units Sold (1,000)") +
  theme_bass() + theme(axis.text.x = element_text(size = 8))
fig_fit
```




```
ggsave("img/fig2_model_fit.png", fig_fit, width = 8, height = 4.5, dpi = 300)
```

```
pred_2023 <- bass.f(14, p, q) * m
data.frame(Reported_2023 = 4210, Predicted_2023 = round(pred_2023))
```

```
##   Reported_2023 Predicted_2023
## 1          4210           5472
```

The data gives 4210 thousand units for 2023, while the model predicts 5,472. This is only a rough check. The model follows the growth well, but it predicts a peak in 2025, while sales actually peaked in 2020. Adding 2023 to the data gives a lower M and a faster model (see the above table). In this case, the estimate should be treated as a rough one.

6 Step 5: Predicting the Diffusion of the i9000

I am using the p and q obtained from the look-alike analysis and will use my estimate of M (see in Step 7). I use the main model (2010-2021).

7 Step 6: Scope

I analyze France, not the whole world, for 3 reasons:

1. The look-alike data describes how the French market behaves.
2. Philips sells the i9000 in France: Check Here
3. GIFAM publishes French market numbers that I will use for a simple Fermi estimate.

8 Step 7: Number of Adopters by Year

M is the number of buyers of the i9000 over its life in France. I estimate it in 3 steps:

> **M = people who own an electric razor x share who buy premium x share of premium that is an i9000**

```
launch_year <- 2025
horizon <- 15 # number of years to predict
pool <- 10.5e6 # electric razors in France, 2024 (GIFAM)

fermi <- data.frame(
  Scenario = c("Low", "Base", "High"),
  Premium_share = c(0.05, 0.08, 0.12), #ASSUMPTION
  i9000_share = c(0.10, 0.20, 0.30) #ASSUMPTION
)

fermi$M <- pool * fermi$Premium_share * fermi$i9000_share
kable(fermi, format.args = list(big.mark = ","))
```

Scenario	Premium_share	i9000_share	M
Low	0.05	0.1	52,500
Base	0.08	0.2	168,000
High	0.12	0.3	378,000

- **10.5 million electric razors** in France in 2024 (GIFAM)
- **Premium share and i9000 share** are assumptions used for Fermi estimation.
- **Check**: the base M is about 3.3% of one year of men's razor and trimmer sales in France (GIFAM)

8.1 Adopters per Year

```
tt <- 1:horizon
years <- launch_year + tt - 1

pred <- data.frame(
  Year = years,
  t = tt,
  a_low = fermi$M[1] * bass.f(tt, p, q),
  a_base = fermi$M[2] * bass.f(tt, p, q),
  a_high = fermi$M[3] * bass.f(tt, p, q),
  A_low  = fermi$M[1] * bass.F(tt, p, q),
  A_base = fermi$M[2] * bass.F(tt, p, q),
  A_high = fermi$M[3] * bass.F(tt, p, q)
)

kable(data.frame(
  Year = as.character(pred$Year),
  Adopters = round(pred$a_base),
  Cumulative = round(pred$A_base),
  Share_of_M = paste0(round(100 * bass.F(tt, p, q)), "%")),
  format.args = list(big.mark = ","))
```

Year	Adopters	Cumulative	Share_of_M
2025	2,169	2,030	1%
2026	2,470	4,347	3%
2027	2,802	6,980	4%
2028	3,162	9,960	6%
2029	3,549	13,314	8%
2030	3,959	17,067	10%
2031	4,386	21,238	13%
2032	4,821	25,842	15%
2033	5,256	30,881	18%
2034	5,676	36,349	22%
2035	6,070	42,225	25%
2036	6,423	48,475	29%
2037	6,720	55,051	33%
2038	6,948	61,892	37%
2039	7,098	68,922	41%

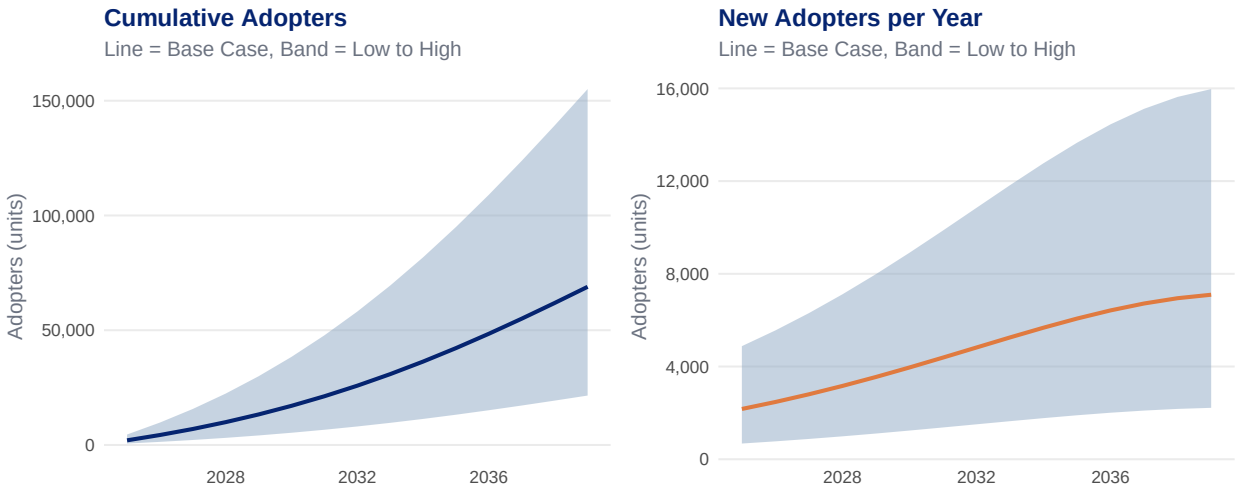
```
data.frame(Peak_year = round(launch_year + t_star - 1))
```

```
##   Peak_year
## 1      2040
```

```
fig_cumul <- ggplot(pred, aes(x = Year)) +
  geom_ribbon(aes(ymin = A_low, ymax = A_high), fill = col_light, alpha = 0.5) +
  geom_line(aes(y = A_base), colour = col_main, linewidth = 1.1) +
  scale_y_continuous(labels = comma) +
  labs(title = "Cumulative Adopters",
       subtitle = "Line = Base Case, Band = Low to High",
       x = NULL, y = "Adopters (units)") + theme_bass()

fig_year <- ggplot(pred, aes(x = Year)) +
  geom_ribbon(aes(ymin = a_low, ymax = a_high), fill = col_light, alpha = 0.5) +
  geom_line(aes(y = a_base), colour = col_accent, linewidth = 1.1) +
  scale_y_continuous(labels = comma) +
  labs(title = "New Adopters per Year",
       subtitle = "Line = Base Case, Band = Low to High",
       x = NULL, y = "Adopters (units)") + theme_bass()

fig_pred <- ggarrange(fig_cumul, fig_year, ncol = 2)
fig_pred
```




```
ggsave("img/fig3_i9000_prediction.png", fig_pred,
       width = 11, height = 4.5, dpi = 300)
```

About 2,169 buyers in 2025, growing to about 7,098 per year in 2039. By then about 68,922 people have bought it. Because q is low, the model puts the sales peak far away.

9 Limits of this Analysis

- The look-alike market already existed in 2010, so p is not a true "launch" value.
- Only 12 data points were used to estimate 3 parameters, so M is highly uncertain.
- Data for 2022 is missing and the 2023 value was an estimate that came with the dataset.
- Sales in 2020 may be higher because of COVID-19 (more grooming at home).
- Razors and trimmers are replaced over time, so M counts units, not people.
- The premium share and the i9000 share in the Fermi estimate are assumptions.